

Intro to AI for Medicine

Lec 10: Practicalities of ML

Last Time

- Discussed the inner workings of LLMs
 - Language embeddings
 - Tokenization
 - Training regime -> self-supervised multiclass classification
 - Attention Block (which is inside the Transformer)
- Real world ramifications of LLMs
 - Environmental concerns
 - Hallucinations and Bias (look to the data!)
- Ways LLMs are innovating in medicine
 - RAGs and document embeddings -> OpenEvidence
 - Multimodality learning and foundation models

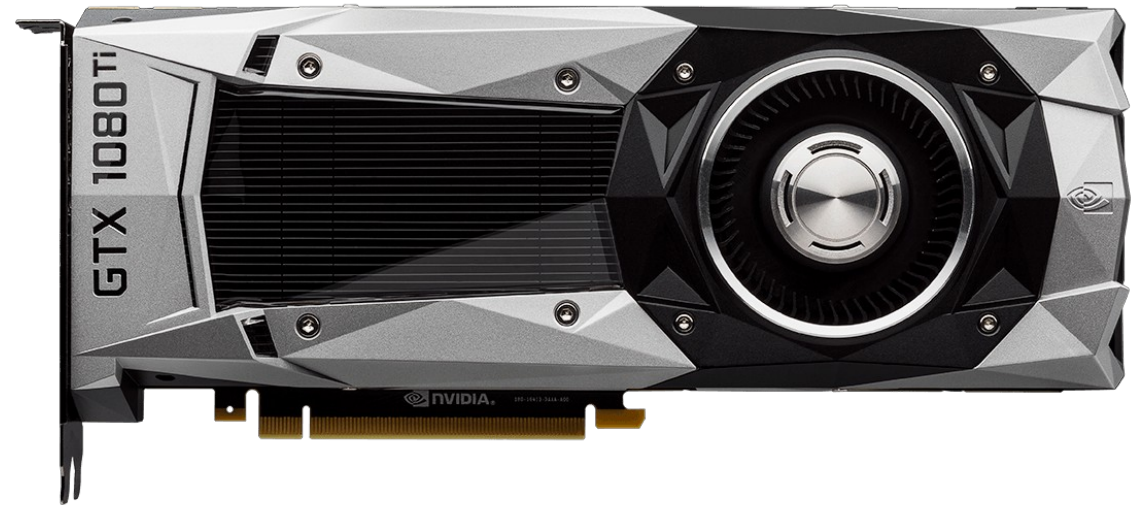
This time

- Where the rubber meets the road: Infrastructure supporting AI/ML
- Out of context learning

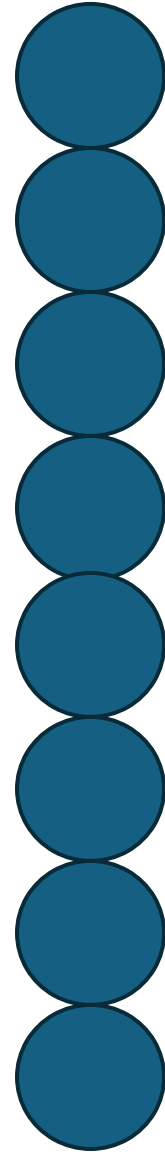


CPU vs GPU

- CPU: Central Processing Unit
- GPU: Graphics Processing Unit



CPU vs GPU



Inside the CPU are “cores”

These cores are assigned operations individually as they come

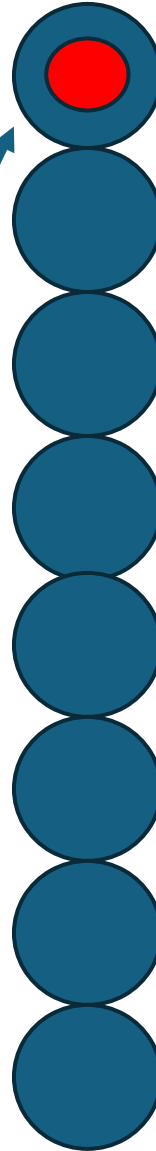
CPU vs GPU

1+1

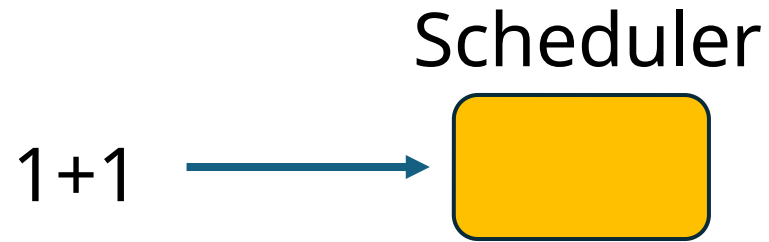
Scheduler

Cores

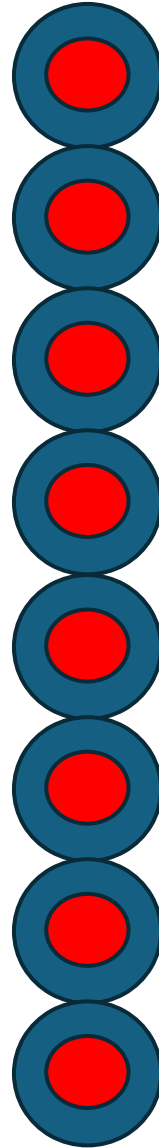
The scheduler assigns to
cores that aren't busy



CPU vs GPU



Cores

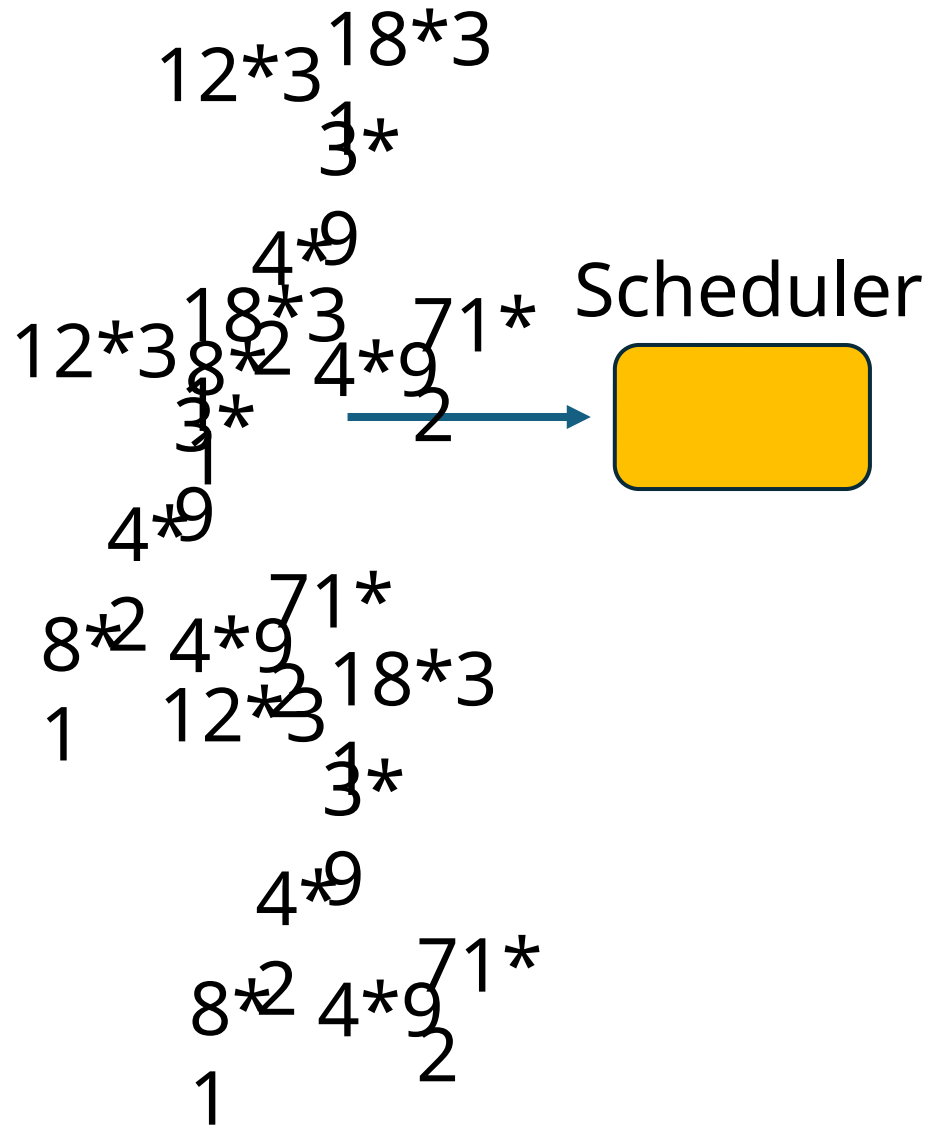


What if all our cores are busy?

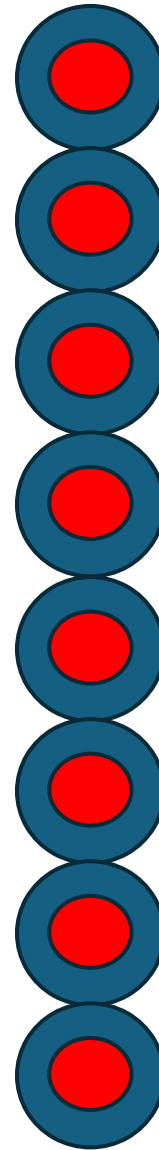
Then we have to wait

Not an issue if we have only a few operations at a time – even if complex

CPU vs GPU



Cores



What if we have a bunch of simple operations?

It would be better if we had a bunch of cores – even if they were less powerful individually

What kind of operation packages a ton of simple multiplications and additions into one?



Matrices: Multiplication

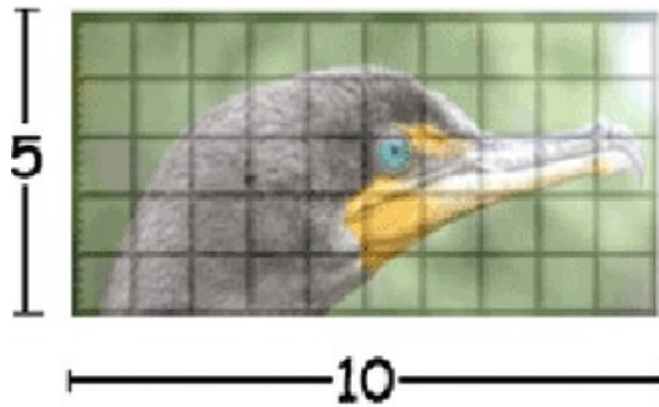
$$\begin{matrix} & A & & B & & C \\ \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} & \begin{bmatrix} b_1 \\ b_4 \\ b_7 \end{bmatrix} & \begin{bmatrix} b_2 & b_3 \\ b_5 & b_6 \\ b_8 & b_9 \end{bmatrix} & = & \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix} \end{matrix}$$

$$a_1 * b_1 + a_2 * b_4 + a_3 * b_7 = c_1$$

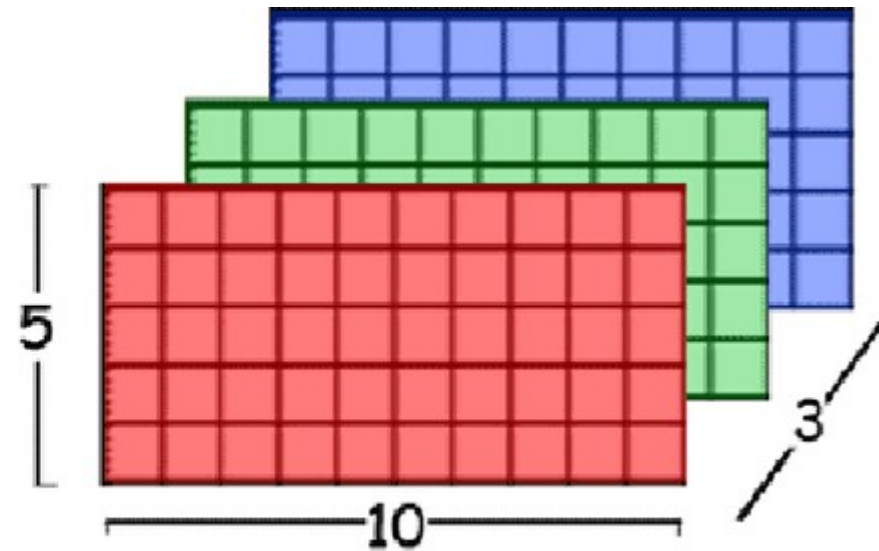
$$[\mathbf{AB}]_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j} = \sum_{r=1}^n a_{i,r}b_{r,j},$$

CPU vs GPU

- Why were GPUs originally used for computer graphics?



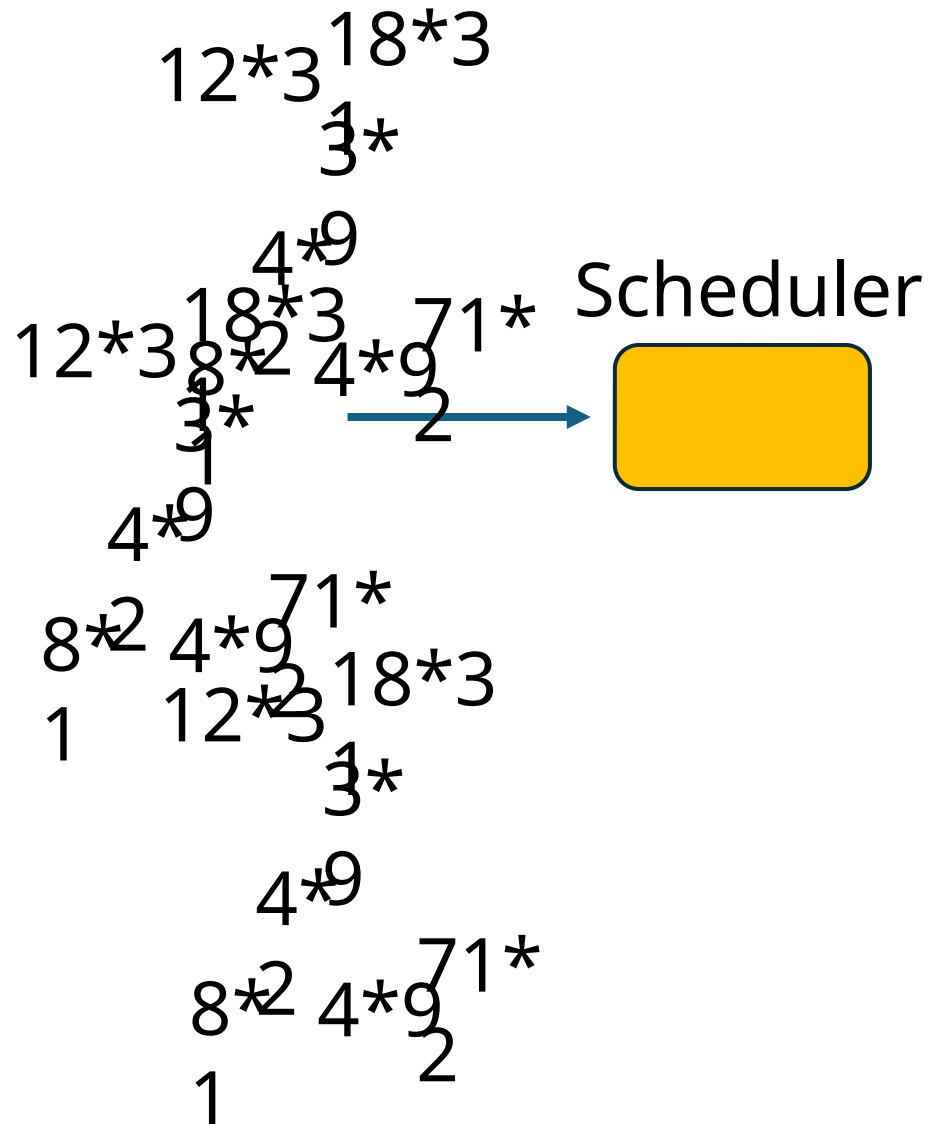
Original Color Image



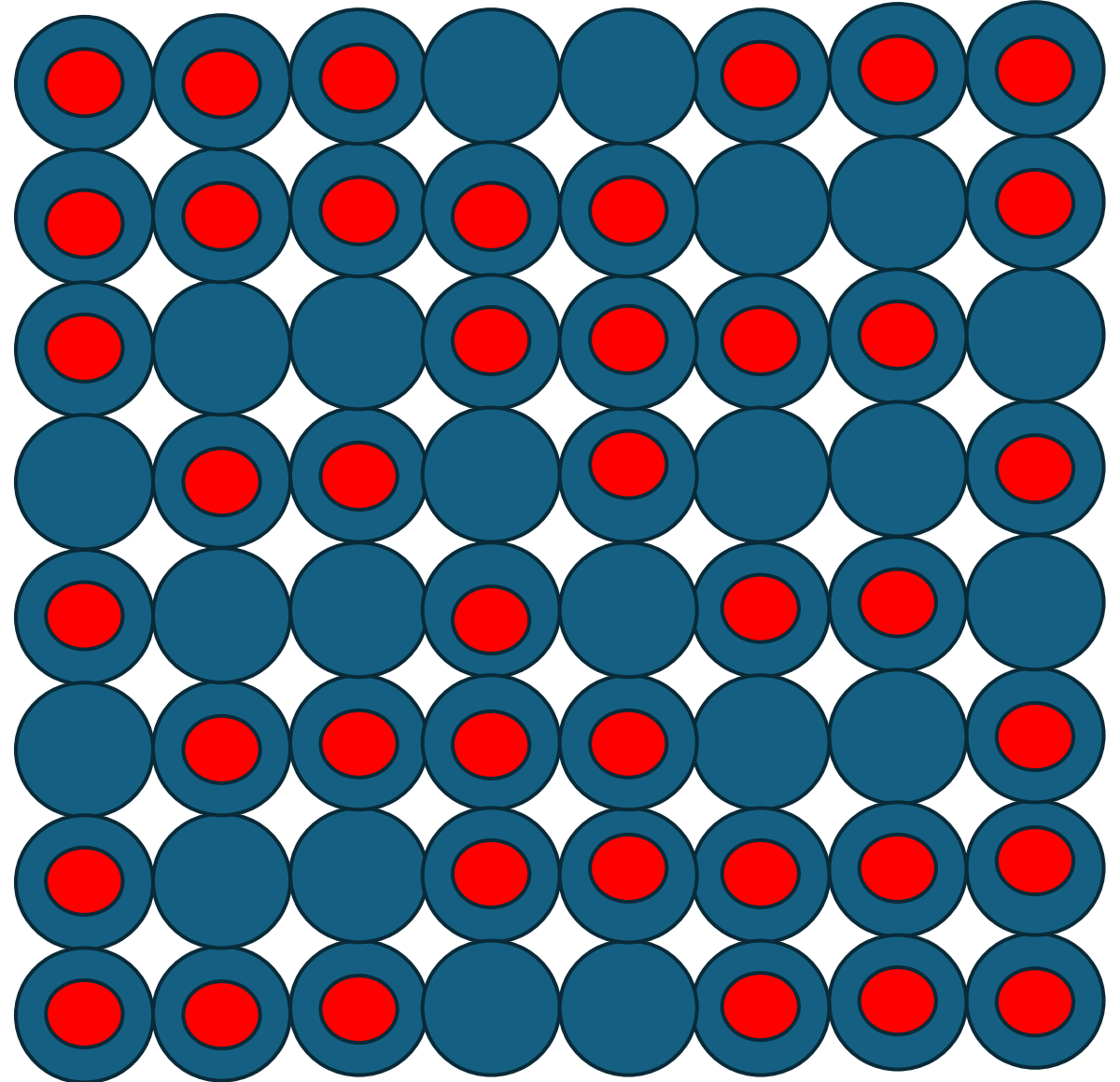
Matlab RGB Matrix

Computer graphics are just another form of matrices!

CPU vs GPU



How GPU cores would look

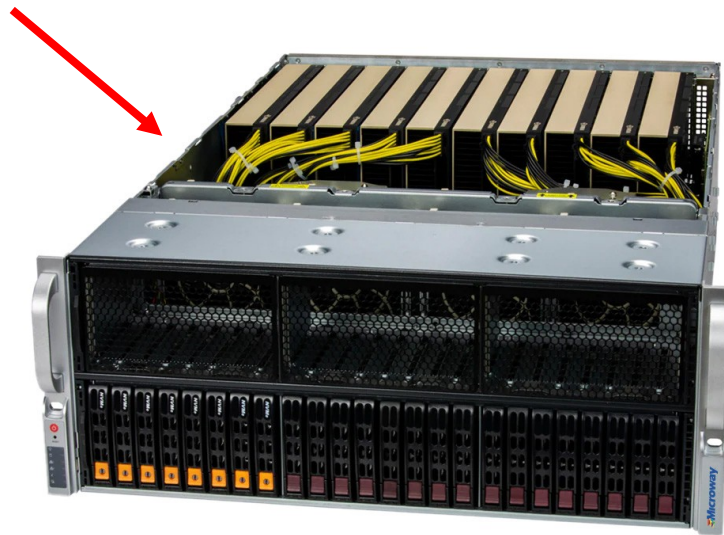


Remember, each is weaker

GPU Takeaways

- Efficient for **highly parallelized** operations, such as matrix multiplication
 - When networks don't play nice with this (RNN), training can take a long time.
 - Stacking layers as tensors instead of separate matrices is faster!
- Even more specialized computers, called TPUs (tensor processing units) are available for ML specific applications
- CPUs are good for operations which must happen one at a time

AI Data Centers



Hypothetical



What are his options for getting the GPUs to execute his idea?

GPU Rack



- Most secure
- Customizable
- Most expensive
- Once you buy you're stuck

Hospital GPU Cluster



- Secure enough
- Less expensive
- Not @ all hospitals
- You get what you get
- Subject to availability

Cloud Compute

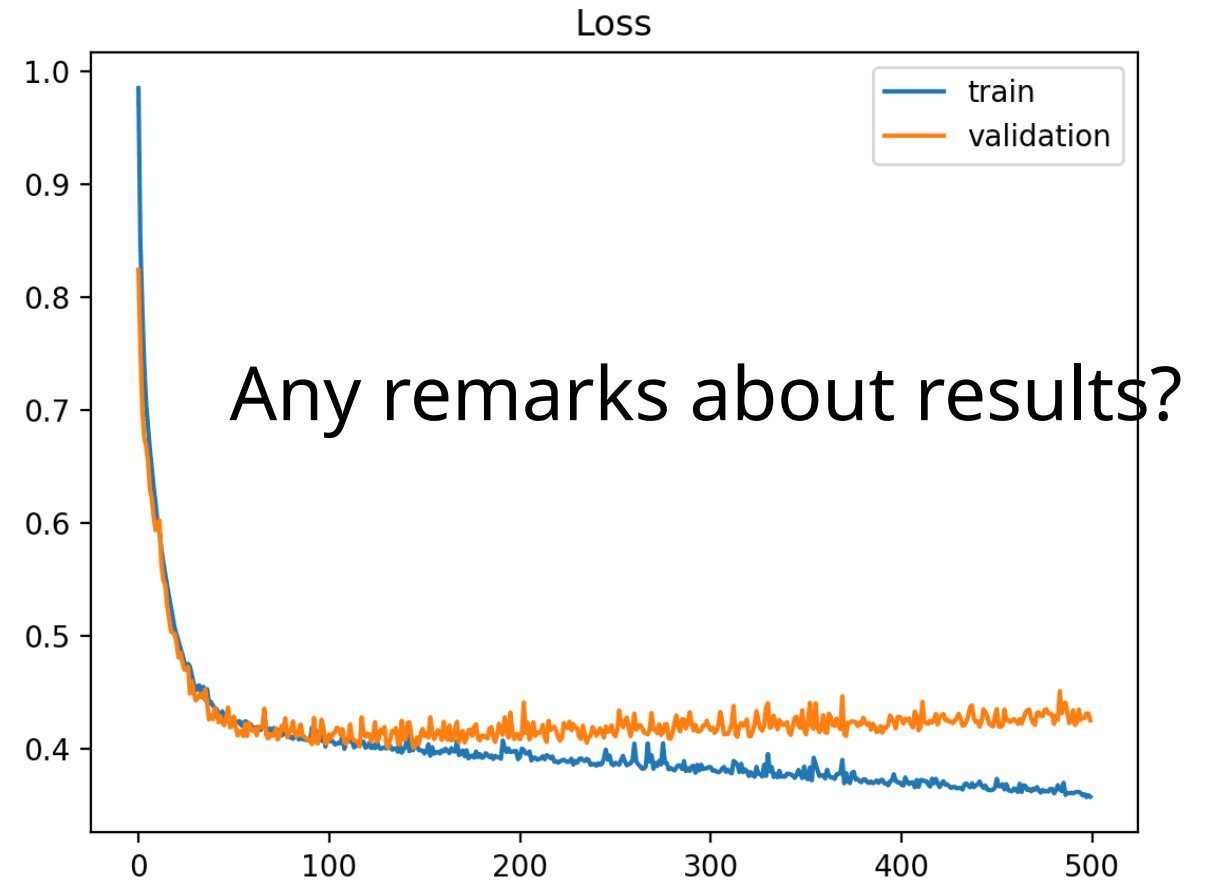


- Lowest barrier to entry
- Plenty of options
- Scales well
- Must meet strict security protocols

What Happened Next?

- Matthew persuades his hospital admin to buy a few GPUs and build a small server cluster
- His team trains a model to predict if a kid will develop severe allergies in their first 5 years of life based on maternal/child data
- He gets the following train/test loss

Preliminary Results



A Collaboration



Federated Learning

Continuing Our Story...

- He decides to publish his paper
- Since it's the CS world, his results can be easily verified
 - Peer review less important

arXiv > cs > arXiv:2510.08498

Search... All fields Search

Help | Advanced Search

Computer Science > Computer Vision and Pattern Recognition

[Submitted on 9 Oct 2025]

A Deep Learning Based Approach to Prediction of Severe Early Childhood Allergy

Matthew Garr, Georgia Garr, Hans Lenhardt, Stephen Lostetter

Early identification of children at risk for severe allergic disease could enable preventative interventions during critical developmental windows. In this study, we developed a deep learning model to predict the likelihood of severe allergy onset by age five using maternal and fetal data collected during pregnancy. The dataset included demographic, clinical, immunologic, and genomic features from 3,200 mother–infant pairs. A multimodal neural network integrated structured data with omics-derived embeddings to capture complex maternal–fetal interactions. The model achieved an area under the receiver operating characteristic curve (AUROC) of 0.87 on an independent test cohort, outperforming traditional logistic regression and random forest baselines. Feature attribution analysis highlighted maternal cytokine profiles and fetal gene expression pathways involved in immune tolerance. These findings suggest that deep learning can uncover biologically meaningful predictors of early allergic disease risk from prenatal data.

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Healthy Skepticism

- Student 1 doesn't believe the results and wants to verify before treating his patients
- Luckily, Matthew's team shared the model weights online

Downloading Model Weights

Prior-Labs/**TabPFN-v2-clf** like 58 Follow Prior Labs 65

Tabular Classification TabPFN License: priorlabs-1-1

Model card Files and versions xet Community 3 Use this model

main TabPFN-v2-clf 281 MB Go to file Ctrl+K 2 contributors History: 21 commits

noahho anuraggarg Upload 5 files (#4) 823e12d VERIFIED about 2 months ago

.DS_Store Safe	6.15 kB		add license	4 months ago
.gitattributes	1.66 kB		Upload tabpfn-v2-classifier-finetuned-z...	3 months ago
LICENSE.txt Safe	11.5 kB		add license	4 months ago
README.md Safe	2.02 kB		Update README.md	4 months ago
config.json Safe	37 Bytes		For Model Count tracking	9 months ago
tabpfn-v2-classifier-finetuned-gn2p4bpt-xp6f0iqb.ckpt	12.9 MB		Upload 5 files (#4)	about 2 months ago
tabpfn-v2-classifier-finetuned-llderlii-oyd7ul21.ckpt	29 MB		Upload 5 files (#4)	about 2 months ago
tabpfn-v2-classifier-finetuned-od3j1g5m-4svepuy5.ckpt	29 MB		Upload 5 files (#4)	about 2 months ago

Datasets

id	sector	occupation	
g · lengths	string · classes	string · classes	
	9 values	44 values	
0b06-26d1-a32c-...	Professional, Scientific, and...	Accountants and Auditors	
d4d-df60-9102-...	Professional, Scientific, and...	Accountants and Auditors	You are the Finance Lead for an advisory client and are responsible for managing and controlling...
9a7-21a7-906f-...	Professional, Scientific, and...	Accountants and Auditors	You are a Senior Staff Accountant at Aurisic. You have been tasked with preparing a detailed...
778-450b-b77e-...	Professional, Scientific, and...	Accountants and Auditors	You are a mid-level Tax Preparer at an accounting firm. You have been given the task to complete an...
943-5a11-b7a2-...	Professional, Scientific, and...	Accountants and Auditors	As our Senior Staff Accountant in Financial Reporting & Assembly, you've been a critical part...
6ac-8f9f-h96c-...	Government	Administrative Services Managers	You are an administrative operations lead in a government department responsible for citizen-

FreedomIntelligence/medical-o1-reasoning-...

Viewer · Updated Apr 22 · 90.1k · 10.1k · 902

robailleo/medical-vision-llm-dataset

Viewer · Updated Sep 2 · 4.79k · 106 · 6

umutertugrul/turkish-hospital-medical-art...

Viewer · Updated 8 days ago · 24.6k · 4.01k · 3

galileo-ai/medical_transcription_40

Viewer · Updated Aug 4, 2022 · 5k · 79 · 22

Malikeh1375/medical-question-answering-da...

Viewer · Updated Nov 1, 2023 · 1.26M · 1.11k · 59

lingshu-medical-mlm/ReasonMed

Viewer · Updated Jun 24 · 1.11M · 474 · 64

Intelligent-Internet/II-Medical-Reasoning...

Viewer · Updated Jul 3 · 2.2M · 190 · 27

moremilk/CoT_Medical_Diagnosis

Viewer · Updated 5 days ago · 3.01k · 30 · 2

keivalya/MedQuad-MedicalQnADataset

Viewer · Updated Oct 11, 2023 · 16.4k · 1.83k · 125

Modalities



3D



Audio



Document



Geospatial



Image



Tabular



Text



Time-series



Video

Size (rows)

< 1K

> 17

Format



json



csv



parquet



imagefolder



soundfolder



webdataset



text



arrow

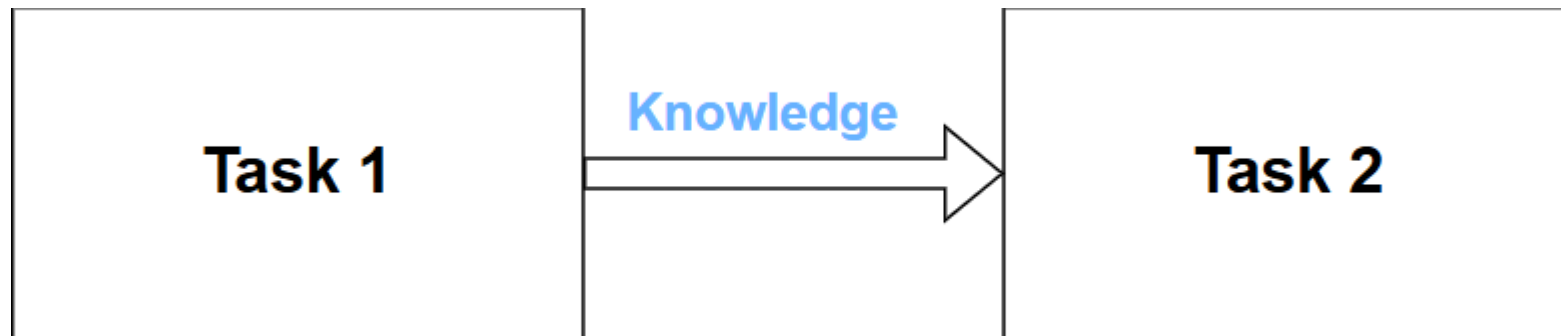
Hypothesis Confirmed



- Parsa downloads the model weights and is able to replicate Matthew & Georgia's results
- Now he has a new clinical question: can the same input predict cardiovascular dysfunction in children

Pretraining & Transfer Learning

- Instead of initializing weights randomly, take them from another task and train a new model
- Any info extracted in hidden layers by original model can be tweaked with GD/Backprop updates
- Pretraining: Training on a much larger, more accessible dataset before applying transfer learning on a smaller, targeted dataset

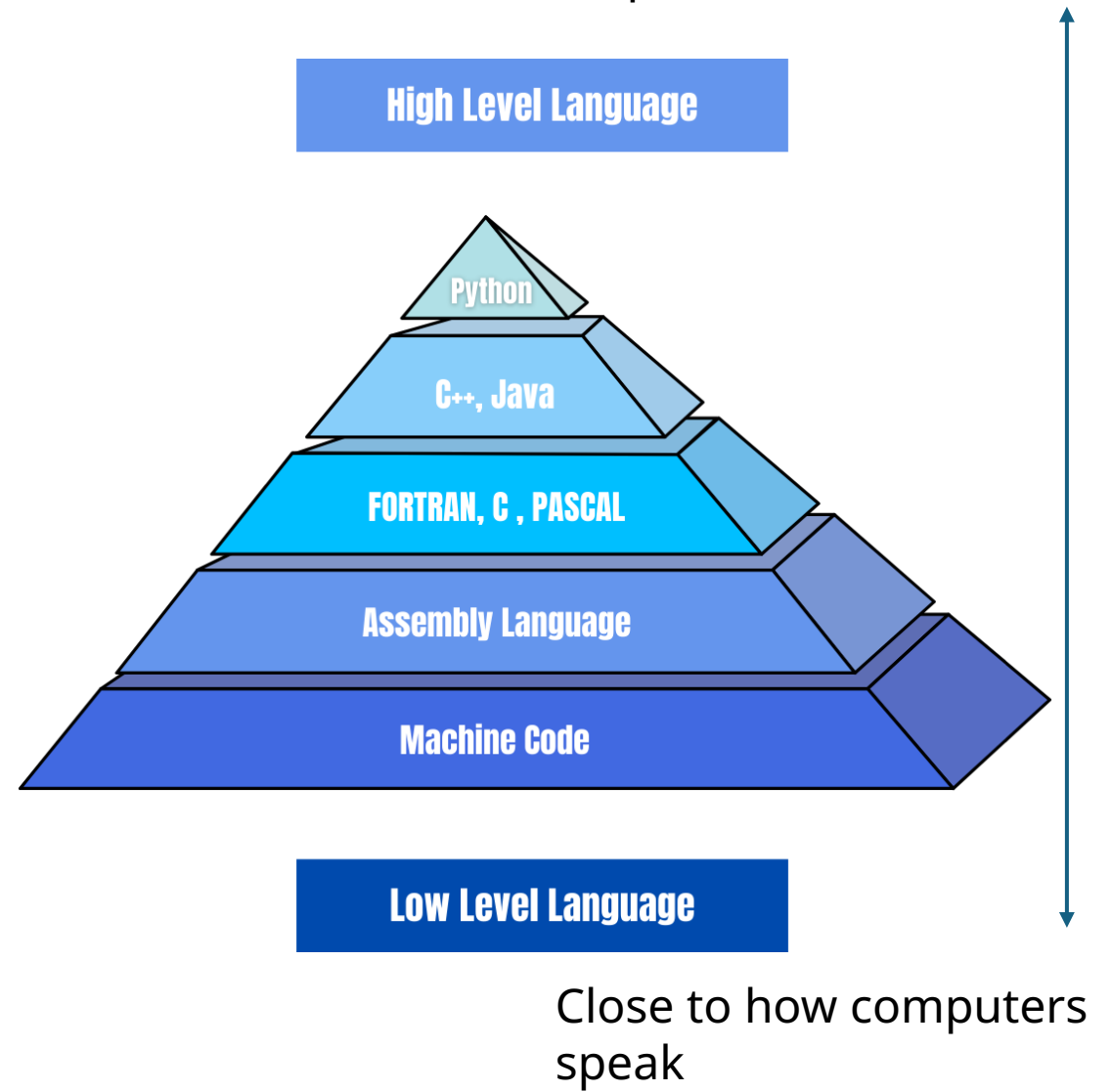


Few-Shot Learning

- Used in cases where labelled data is scarce
 - Rare disease
 - Image segmentation
- N-way, K-shot learning
 - N = number of classes
 - K = number of training examples
 - So if 3 way 2 shot, three classes with 2 labelled examples each

CS Stuff: Programming Languages

- Over time, programming has become “abstracted” or closer to how humans speak
- Under the hood is code to translate between levels
- Code closer to how computers speak requires less translation and therefore runs faster
- All AI coding is done in Python



Machine Learning Python Packages

- Combines multiple operations together and runs them in low level language to speed up AI
- Not relevant for medical practice but good to know



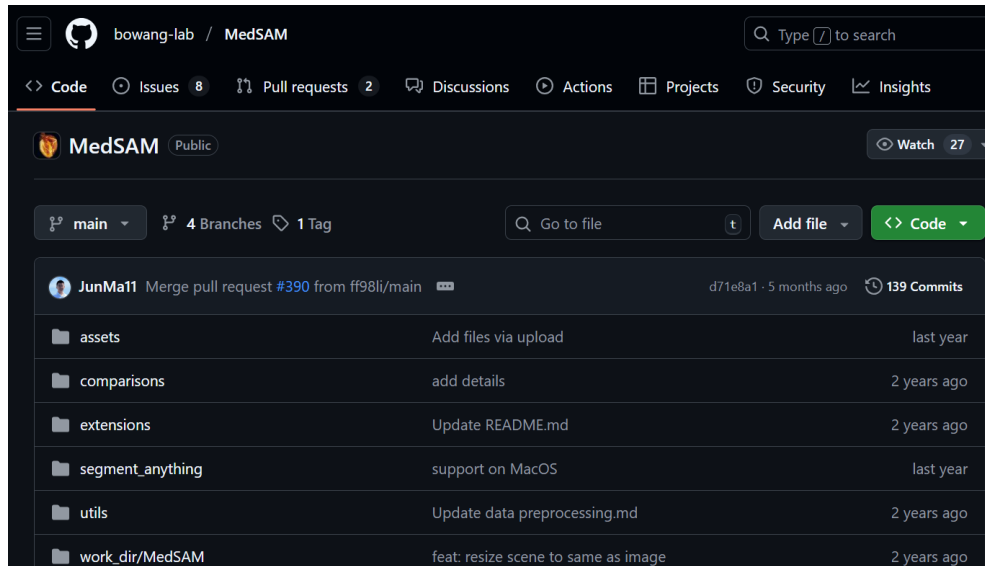
```
for epoch in range(EPOCHS):  
    print('EPOCH {}'.format(epoch_number + 1))  
  
    # Make sure gradient tracking is on, and do a pass  
    model.train(True)  
    avg_loss = train_one_epoch(epoch_number, writer)
```

Building a team

- Clinician: identify clinical pain point or question
- Data engineer: source and clean data
- Data scientist: perform analysis on data, including running models
- Software engineer: build consistent applications that house models
- Clinical Ethicist: Act as consultant regarding bias/safety of model

The Final Product

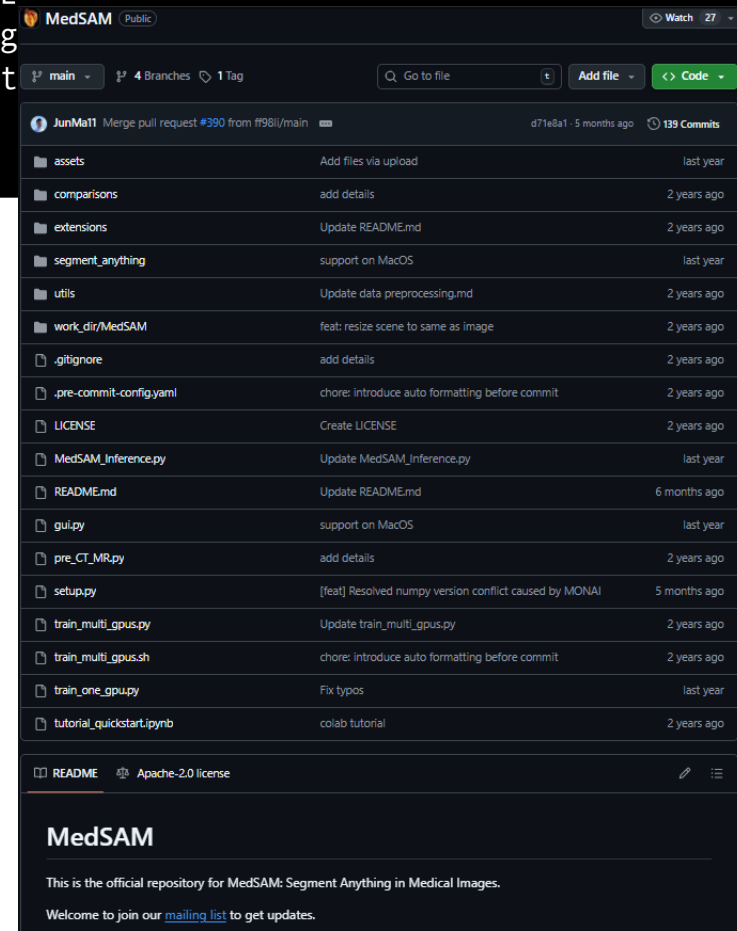
- The whole team works together with their software engineers to build & package a program for others to use



Git & Github

- Git: version control software so many engineers can work on software independently and later sync
- Github: Hosting site for these versions
- Github contains lots of open source coding projects which can be pulled and repurposed by your engineers

```
$ git init
Initialized empty Git repository in /tmp/tmp.IMBYSY7R8Y/.git/
$ cat > README << 'EOF'
> Git is a distributed revision control system.
> EOF
$ git add README
$ git commit
[master (root-commit) e4dcc69] You can edit locally and push
to any remote.
 1 file changed, 1 insertion(+)
 create mode 100644 README
$ git remote add origin g
$ git push -u origin mast
```

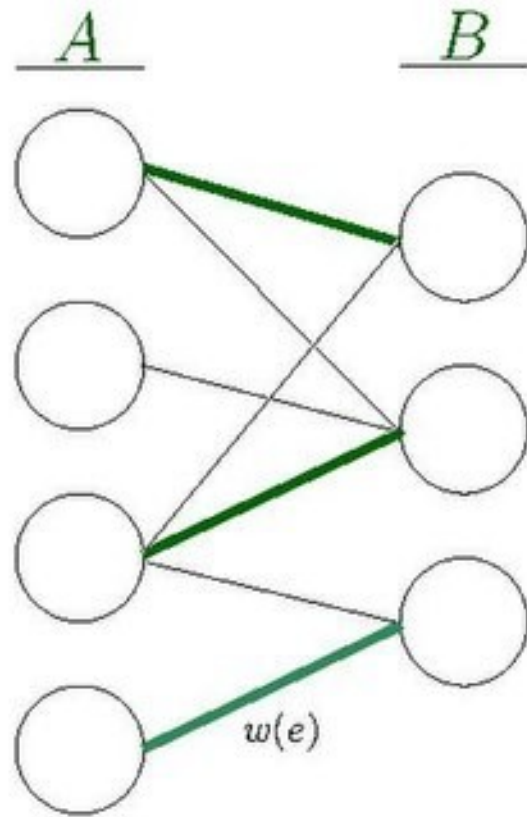


A word on regular algorithms

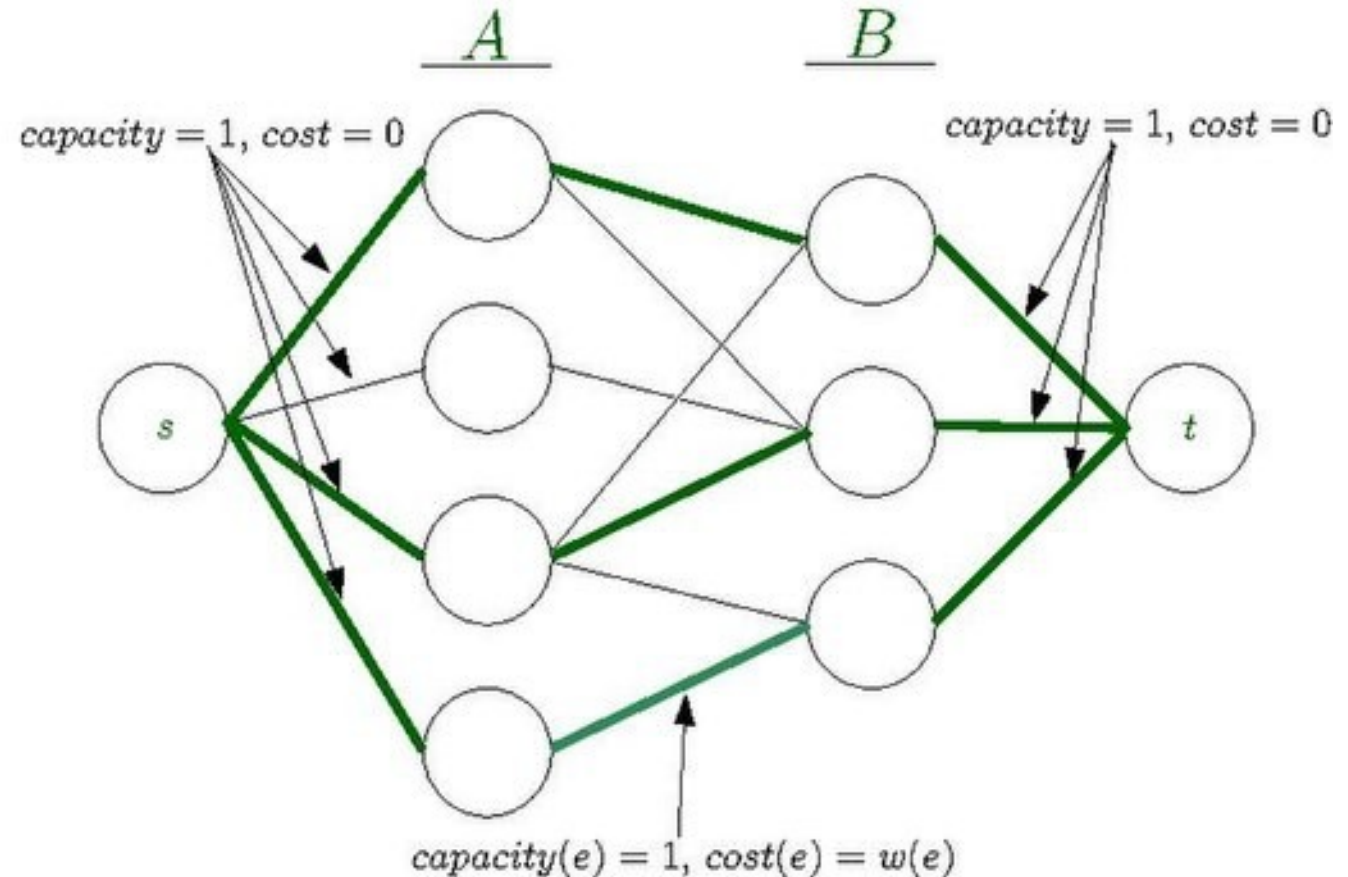
AI Algorithms	Regular Algorithms
Have randomness involved (can get caught in local minimum) Stochastic	Always find same answer, every time (usually) Deterministic
Take in an input and output and find an algorithm mapping them	Takes in an input and algorithm and returns an output
Algorithm returned is always an approximation	Algorithm usually is perfectly precise
Advanced techniques needed to explain decision making	Choices made completely explainable

UNC SOM Campus Match

$G :$



$G' :$



Next time...

- Discussing challenges with rolling out AI in clinical contexts

Thanks!

